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Adjustable AC CIRCUIT BREAKER

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This circuit breaker employs a single operational amplifier (op-amp) and yet has a wide range and is user-friendly. A circuit breaker is an electrical switch intended to protect an electrical circuit or device from damage caused by excess current flow or short-circuit. A basic circuit breaker includes a simple fuse and a normal miniature circuit breaker (MCB). It is designed for a fixed-rated current value in terms of amperes.

In this circuit, you can select the current value range from milliamperes (mA) to a few amperes. A simple circuit compares current with a preset value. If current increases beyond the preset value, the circuit breaks (opens) the current flow to the load. The circuit then automatically closes and restores after a specified time or manual reset. The block diagram of the circuit breaker is shown in Fig. 1.

Circuit and working

The circuit diagram of the adjustable AC circuit breaker is shown in Fig. 2. It is built around transformer X1, six 1N4007 diodes (D1 through D6), one 741 op-amp (IC1), two LEDs (LED1 through LED2), two npn transistors BC548 (T1 and T2), 15V Zener diode ZD1, 12V-1CO relay and a few other components.

The circuit is constructed around the 741 op-amp. The op-amp is used in non-inverting comparator mode.

Major construction lies in the design of the current transformer as explained below. Take 230V AC primary to 9V-0-9V, 1A secondary transformer (X1) as shown in Fig. 3. In this transformer, usually low-voltage secondary winding is wound around 230V AC primary winding. X1 has 230V AC primary (say, 230V AC side) and 9V-0-9V, 1A secondary, but here secondary winding is unused (need not be removed). A few turns (three to five) of 20SWG enamelled copper wire, say L1, is wound on top of the winding (around its bobbin), as shown in Fig. 4.

Switch S1 is an on/bypass switch. You can bypass the circuit breaker system by keeping S1 towards position 2 (bypass), where load current flows directly to AC output through fuse F1. Here, F1 acts as a traditional protecting device.

If S1 is at position 1 (on), load current flows through L1 winding of X1 and relay RL1 contacts, to AC output connected to AC load.

Load current is passed through L1, and is used for load current measurement. Voltage produced at 230V AC side of the transformer is proportional to the amount of load current passing though L1 winding of X1. Voltages corresponding to the load current flowing through winding L1 are listed in Table I. Values may vary depending on number of turns and value of R2.

Output of 230V AC side is rectified using a bridge rectifier. Resistor R2 limits the voltage developed at 230V AC side of X1. Rectified voltage is filtered and connected to R3 and R4.

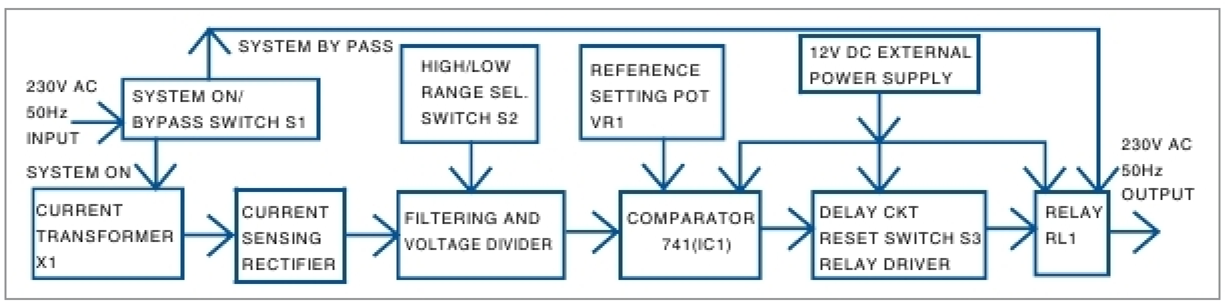

Fig. 1: Block diagram of adjustable AC circuit breaker

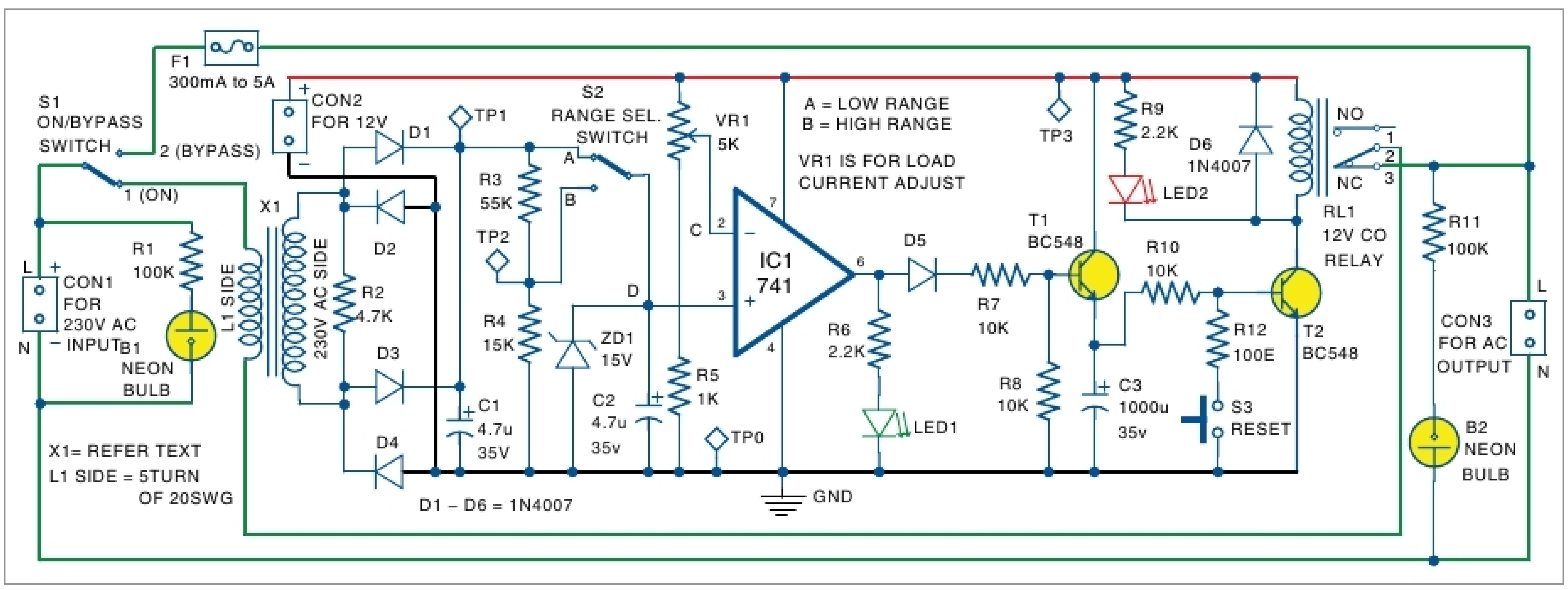

Fig. 2: Circuit diagram of adjustable AC circuit breaker